

The digital age has brought about an unprecedented transformation in almost every aspect of society. From the expansion of data centers to the rise of artificial intelligence (AI) and cryptocurrencies, these technologies have reshaped industries and everyday life. However, this rapid advancement comes with significant environmental challenges, primarily due to carbon emissions. As data centers, AI systems, and cryptocurrency operations consume vast amounts of energy and resources, the resulting carbon footprint has become a growing concern. In this essay, we will explore five major problems related to carbon emissions in digital technologies and propose detailed solutions for mitigating their environmental impact.

Firstly, we will estimate the carbon emissions of digital technologies, such as data centers, AI, and cryptocurrencies, which create significant environmental impacts. We develop a model divided into two main components: the Operational Carbon Footprint and the Infrastructure Carbon Footprint. The operational carbon footprint refers to emissions generated by energy usage during the operation of systems like data centers, AI, and cryptocurrency operations. It includes the energy consumed by servers, cooling systems, and other infrastructure used during daily operations. The infrastructure carbon footprint encompasses emissions generated by the construction and maintenance of physical infrastructure, such as data centers. This includes emissions from building materials (e.g., concrete and steel) and the associated processes that require energy, often from fossil fuels. HPC carbon emissions in 2030 are predicted to be 339.66 million tons.

As digital technologies continue to grow and evolve, their energy consumption and associated emissions are expected to rise. It is crucial to predict how emissions will change in the future, taking into account factors such as technological advancements, policy changes, and energy market fluctuations. Therefore, we use time-series analysis to predict future emissions. Time-series models analyze historical data on energy consumption and carbon emissions, extrapolating this data into the future while adjusting for anticipated changes in energy mix, technological advancements, and global policies. This model also accounts for the expected growth of digital technologies (data centers, AI, and cryptocurrencies) and predicts how this growth will impact overall emissions.

The energy mix powering digital systems significantly affects their carbon emissions. Fossil fuel-based energy sources like coal and natural gas are carbon-intensive, whereas renewable energy sources like wind, solar, and hydro are much cleaner in terms of emissions. We build a model to quantify the impact of energy sources on carbon emissions. By analyzing energy consumption data, we can determine how much CO_2 is emitted based on the types of energy used. As countries and companies shift toward renewable energy, the carbon intensity of energy consumption decreases, helping to reduce the overall carbon footprint of digital technologies. The model also predicts how the transition to renewable energy will influence land consumption and costs. The results suggest that as renewable energy becomes more prevalent, the operational carbon footprint of digital technologies will decline.

In addition, we aim to determine the optimal placement of data centers within a network to balance construction costs and latency while ensuring efficient data transmission. The network is represented as a graph where nodes represent locations where data centers can be placed, and edges represent connections between nodes, with distances indicating latency. We use Integer Linear Programming to minimize a cost function that combines latency and construction costs. Then, we apply Dijkstra's Algorithm to determine the shortest paths for data transmission between nodes, optimizing routing to minimize energy and time. This approach can also be applied to site selection to reduce the carbon footprint. We modulated the preference for priority emission reduction or latency optimization to find the most effective locations, as represented by a graph in our essay. As government subsidies increase, carbon dioxide emissions from network traffic will decrease.

Finally, we develop practical recommendations to lessen the environmental impact of high-performance computing. We conduct thorough simulations of one of our strategies and demonstrate its ability to cut carbon emissions and lower energy usage.

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1 Introduction

1.1 Background

High-Powered Computing (HPC), also known as high-performance computing, refers to the use of supercomputers and parallel processing techniques to solve computational problems at high speeds. There is a growing reliance on HPC for advanced technologies, such as artificial intelligence, data science, and cryptocurrency mining. However, HPC's high energy demand primarily relies on fossil fuel-based electricity, contributing to excessive carbon emissions, which directly accelerate climate change. The massive environmental impact can be illustrated by an academic finding: training a single deep learning model can emit over 284 metric tons of carbon dioxide, comparable to the lifetime carbon emissions of five cars. Additionally, data centers place considerable stress on local power grids, particularly in areas with limited renewable energy infrastructure, further increasing dependency on nonrenewable sources.

Moreover, HPC systems process vast amounts of data, requiring high-throughput networks to transfer large datasets between processors, storage systems, and peripheral devices. These high-speed connections with low latency consume significant power, adding to the overall carbon footprint. However, such operations are not sustained constantly. In this context, average utilization rates provide a more realistic view of annual energy consumption, as HPC systems are often underutilized.

We can describe this problem in terms of the annual energy consumption of HPC capabilities, considering both full capacity and average utilization rates. Full capacity refers to the energy usage when HPC systems operate at maximum computational load. This metric provides a baseline for the peak energy demands of HPC infrastructure worldwide, highlighting the potential maximum impact on global energy resources. It can be calculated by summing the total power requirements of all HPC systems and data centers globally when running at peak performance.

However, HPC systems rarely operate at full capacity constantly. In this case, average utilization rates offer a more realistic perspective on annual energy consumption. Factors such as job scheduling, task types, and processing downtime contribute to these variations, making average utilization rates crucial for estimating day-to-day energy needs. According to IBM, a world-renowned artificial intelligence solutions and cloud platform company, the average server utilization rate is only 12–18% of capacity. This creates opportunities to leverage cloud computing and virtualization technologies to optimize resource allocation. By dynamically scaling workloads based on actual demand, these technologies can significantly reduce energy waste and carbon emissions. Such optimization aligns with subsequent discussions on the facility location problem, enabling more efficient system designs, which we will explore later in the essay.

Evaluating the environmental impact of high-performance computing (HPC) is essential for advancing sustainable development goals, particularly those related to climate action and clean energy. By assessing HPC's carbon footprint and energy requirements, we can gain critical insights into areas where energy efficiency can be improved, renewable power adoption can be increased, and resource usage can be optimized. These evaluations align with the United Nations' Sustainable Development Goals (SDGs), which emphasize reducing greenhouse gas emissions, promoting sustainable infrastructure, and ensuring that technological advancements do not exacerbate environmental harm. Assessing HPC's environmental impact not only supports these global goals but also drives innovation in green computing practices, ensuring that the growth of HPC technologies remains environmentally responsible and sustainable for future generations.

1.2 Problem Restatement

Our goal is to examine the environmental impact of High-Powered Computing (HPC), focusing specifically on its global energy consumption and associated carbon emissions. We aim to address multiple interconnected problems to quantify, analyze, and propose solutions for mitigating the environmental impact of HPC, both in the present and the future.

Problem 1 We will first construct a detailed model to estimate the carbon emissions resulting from HPC operations. This model will account for diverse energy sources, examining their respective energy mixes and carbon intensities. By integrating these variables, we aim to provide a comprehensive understanding of how energy choices contribute to the carbon emissions caused by HPC.

Problem 2 Building upon the model from Problem 1, we will assess the potential evolution of HPC's

environmental impact. This includes analyzing the expected growth of HPC, the increasing energy demands from other sectors, and possible shifts in energy sourcing strategies. We will particularly forecast how these factors might shape HPC's energy consumption and emission profile by the year 2030.

Problem 3 To deepen the analysis, we will expand the model to evaluate potential reductions in carbon emissions resulting from the full adoption of renewable energy. We will explore the effects of transitioning to renewable energy at different scales. Our goal is to quantify the potential emissions reductions achievable through such shifts and provide a framework for assessing the environmental benefits of renewable energy adoption.

Problem 4 Given the significant impact of network architecture on the energy consumption of HPC, we will design a dynamic network architecture that can scale up or down based on real-time demand. This approach aims to optimize resource usage and reduce HPC's carbon emissions.

Problem 5 Building on our earlier research, we will develop practical recommendations to lessen the environmental impact of HPC. We will conduct thorough simulations on one of these strategies to demonstrate its ability to reduce carbon emissions and lower energy consumption effectively.

2 Assumptions and Variables

2.1 Assumptions and Justifications

Several assumptions are listed to simplify the problem. The assumptions and their justifications are listed below.

Assumption 1. When choosing the site of data centers, enterprises only consider the construction cost and service quality (which can be indicated by latency).

Justification Construction cost and service quality are the most immediate and quantifiable factors influencing site decisions. While other considerations, such as environmental impact or energy efficiency, often take a backseat to cost-effectiveness and user experience. We assume that the position of data centers is only affected by this two factors, allowing a focused analysis of key trade-off.

Assumption 2. There is no effect on infrastructure carbon footprint due to technological progress.

Justification While technological progress can reduce per-unit emissions, its impact on the overall infrastructure carbon footprint is often offset by increased demand and slow adoption rates. The long lifespan of infrastructure and the delayed effects of technological changes making it appropriate to assume the carbon footprint will remain at the same level during period of scientific progressing.

Assumption 3. When considering total energy usage of HPC, we only count the energy usage of data center and cryptocurrency of Bitcoin, which are the two major factors counted in energy consumption.

Justification Data centers are energy-intensive due to the computational and cooling demands, while cryptocurrency mining has surged in energy usage in recent years due to its reliance on large-scale computational power. We exclude other potential energy-consuming activities, such as some smaller-scale operations, to simplify the model and focus on primary drivers.

Assumption 4. There is no reduction in carbon emission due to the use of green building material.

Justification While green materials may lower emissions in theory, their real-world adoption often faces challenges, such as higher costs, limited availability, and inconsistent performance across regions. Additionally, the production of these materials may still involve carbon-intensive processes, offsetting potential reductions of carbon emission. Therefore, we assume there is no reduction of carbon footprint to allow the model focus on other key factors to ensure its applicability to current construction practices.

Assumption 5. The emissions of CO₂ only take into account the increased energy consumption caused by traffic and local energy mix.

Justification Traffic-induced energy consumption and the local energy mix are primary factors influencing emissions, as they can directly affect the demand for energy and its associated carbon intensity. Other potential contributors, such as embodied emissions from construction or maintenance are excluded to avoid introducing variables with high uncertainty or less immediate relevance. We tend to focus on key factors that are both measurable and impactful in the short term.

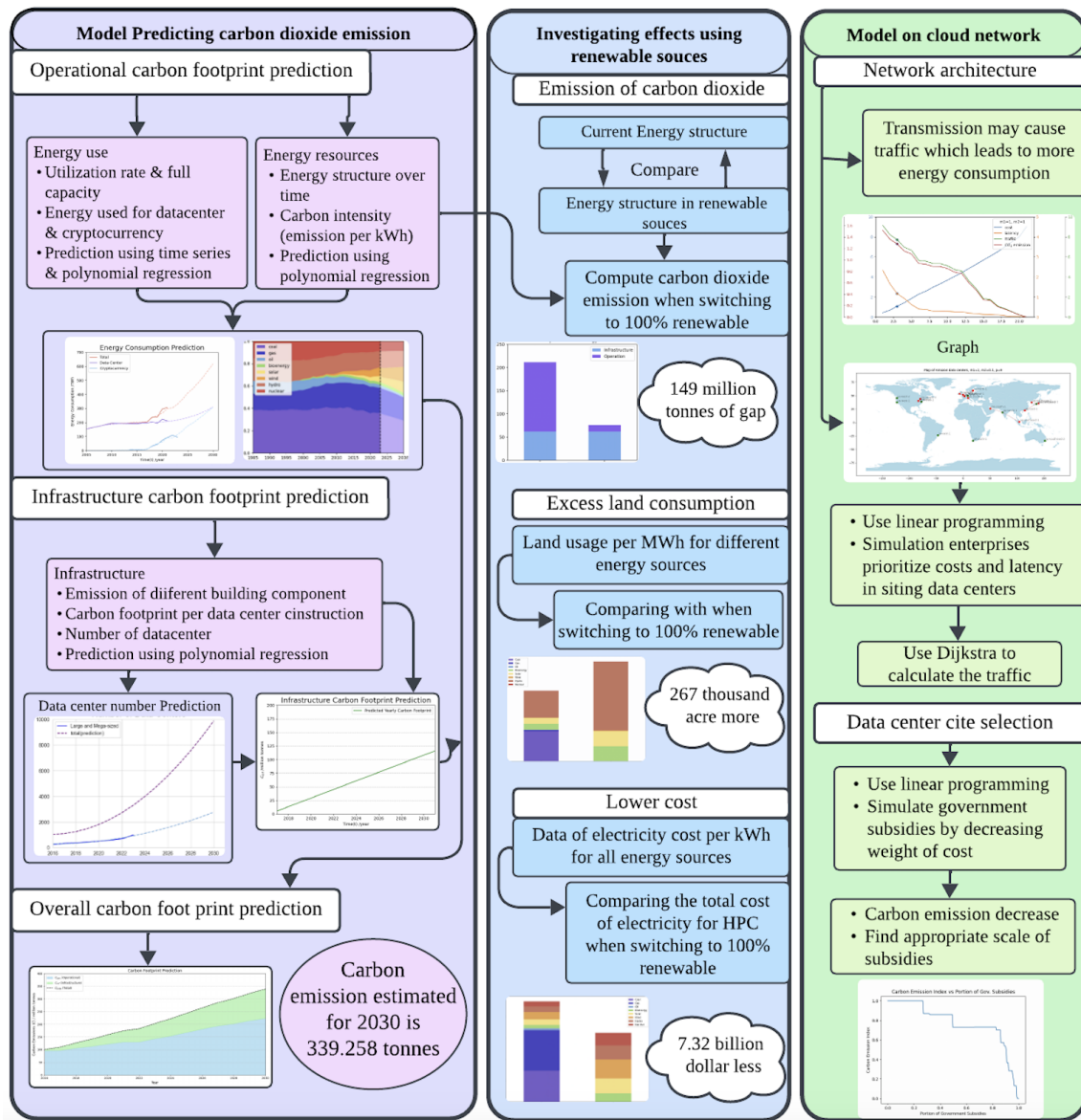


Figure 1: Our modeling process

2.2 Our work

Our modeling process is shown as in Figure 1.

In our modeling process of the first problem that predicts carbon dioxide emission, we approach it by splitting it into operational and infrastructure carbon footprints. For operational carbon, we used polynomial regression to predict the energy use in HPC (data center and cryptocurrency) and energy structure over time and searched for data on carbon intensity and utilization rate. Moving on to infrastructure carbon, we estimate the construction pollution of the data center as 40 times more than an office building and use polynomial regression to predict the data center number and infrastructure carbon footprint over time. Overall, this model integrates these predictions to calculate the overall carbon footprint of carbon dioxide emissions for 2030.

Next, in the second problem that investigates the effect when energy resources are 100% renewable, we start with carbon dioxide emission which is expected to increase. We calculate the exact numerical value by comparing the two energy structures and applying carbon intensity, then analyze the result and provide suggestions for the best energy. Secondly, we compute and analyze the excess land requirement for renewable energy by applying land usage efficiency data. Thirdly, we discovered that the cost of energy in renewable energy is lower than normal by finding each of them through data on energy prices. Overall, we suggest nuclear and wind as the best energy source, because they cost less land and money.

2.3 Variable and Definitions

The variables are showed in Table 1.

Variable Symbol	Definitions
C	Overall g global carbon dioxide emission
C_{op}	Operational carbon footprint
C_{inf}	Infrastructure car
E	Energy demand
$e.t.$	Energy type
c	CO_2 intensity of energy resources (emission per kWh)
p	Proportion of an energy source to total energy
N	Number of newly constructed data center every year
m_1, m_2	weight of latency and cost in objective function
X_i	DC on node i
Y_{ij}	node i serve for node j
P	increment of data centers
L_i	latency between node i and j
D_j	demand of node j
K_i	cost of building DC on node i
F	facilities set
i, j	facility index

Table 1: Variables table

3 Carbon dioxide emission Prediction

3.1 Model overview

We build a model to explore and predict carbon dioxide emissions. In the model, we describe total carbon dioxide pollution as the sum of emissions created during the construction and maintenance of facilities and devices in data centers, as well as the indirect pollution due to the rise in electricity demand for data centers, AI, and cryptocurrencies. This model comprehensively accounts for emissions during both the construction and maintenance phases, alongside indirect emissions from increased energy demand.

We divide the total carbon footprint into operational and infrastructure carbon footprints, exploring them separately. First, we calculate the total energy usage by summing the energy consumption of data centers and cryptocurrencies. Since total energy consumption is distributed across various energy sources, we calculate the energy consumed from each source based on its proportion. Each energy source emits a specific amount of carbon dioxide per unit of energy, enabling us to calculate emissions for each source. Finally, we sum emissions from all energy sources to determine the operational carbon footprint. For the infrastructure carbon footprint, we calculate carbon emissions based on the occupied area of infrastructure. By estimating the number of newly constructed data centers each year, we build a model to predict the infrastructure carbon footprint until 2030. Consequently, we can derive the total predicted carbon footprint.

3.2 Carbon Emission

We denote the total carbon footprint, the operational carbon footprint and the infrastructure carbon footprint as, C , C_{op} , C_{inf} respectively. We can calculate the carbon footprint of global high performance computing as the sum of operational carbon footprint C_{op} and infrastructure carbon footprint C .

$$C = C_{op} + C_{inf} \quad (1)$$

3.3 Operational Carbon Footprint

The Operational Carbon Footprint represents emissions caused by energy usage during system operation. It includes emissions from energy consumption required to power facilities, devices, and processes such as data centers, artificial intelligence (AI), and cryptocurrency operations.

This footprint highlights the direct environmental impact of running energy-intensive systems:

$$C_{op} = \sum E_{e.t.} \cdot c_{e.t.} \quad (2)$$

In our model, we calculate the total operational carbon footprint by the summation of carbon emissions of multiple energy sources. While $E_{e.t.}$ represents the energy consumption of a particular energy type. For example, a portion of the total electricity is generated by coal, while another portion is produced by natural gas. We use $c_{e.t.}$ to indicate the amount of carbon dioxide emitted per kWh of a specific type of energy source consumed. We will consider that different energy sources have varying $c_{e.t.}$.

3.3.1 Energy usage

We investigate the energy usage of data centers and cryptocurrency (Bitcoin) in high-performance computing (HPC), as they are the primary sources of electricity demand. The total energy consumed by HPC is the sum of data center and cryptocurrency energy consumption: The total energy consumed by HPC is, in Equation 3, the sum of data center and cryptocurrency.

$$E_{op} = E_{data\ center} + E_{cryptocurrency} \quad (3)$$

First, we predict energy consumption in cryptocurrency activities by first researching cryptocurrency consumption index, then build a time series model for data processing. The result is shown in Figure 2. [4]

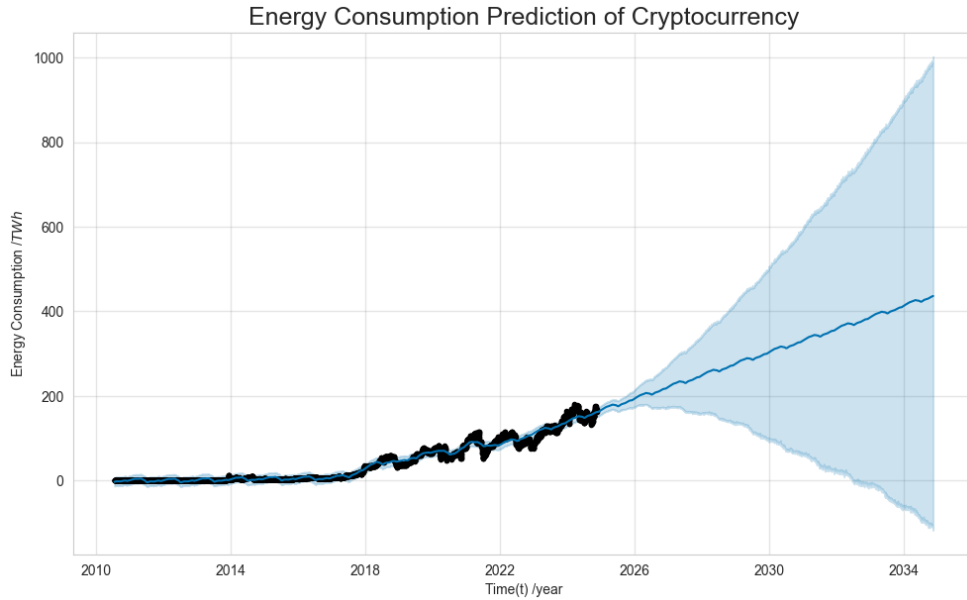


Figure 2: Energy Consumption of Cryptocurrency

As shown in Figure 2, we utilized the ARIMA model to forecast cryptocurrency energy consumption based on historical data. The black dotted line represents observed historical data, the blue solid line indicates the predicted trend, and the blue shaded area represents the uncertainty range (95% confidence interval). The figure demonstrates a continuous upward trend in energy consumption.

The ARIMA models, or the auto-regressive moving average model, can recognize a set of different standard time structures in time series data. The ARIMA model can separate the noise from the signal and then projects the signal into the future to make predictions, and is suitable for fitting data showing non-stationarity. The ARIMA model has three parameters: p , d and g , where p is used to represent the lag number of time series data itself, d represents the time series data which need to undergo several orders of difference transformation to reach the most stable level, and g represents the lag number of prediction errors adopted in the prediction model. In the case of known parameters. the mathematical representation of ARIMA is:

$$AR(X_t, p) = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \phi_3 X_{t-3} + \cdots + \phi_p X_{t-p} \quad (4)$$

$$MR(X_t, q) = e_t + \theta_1 e_{t-1} + \theta_2 e_{t-2} + \theta_3 e_{t-3} + \cdots + \theta_q e_{t-q} \quad (5)$$

The ARIMA model shows a good fit with historical data, at the same time the uncertainty of the predictions increases over time. Similarly, we use a polynomial regression model to predict the total energy consumption of data centers each year from 2024 to 2030, as shown in Figure 3. [1]

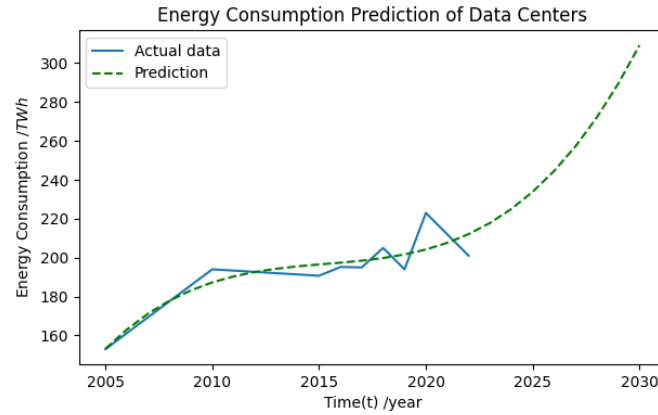


Figure 3: Energy Consumption of Data Centers

In Figure 3, the blue solid line represents the actual energy consumption data from 2005 to 2024, while the green dashed line shows the predicted energy consumption trend extending to 2030.

The historical data indicates a generally increasing trend, with some fluctuations. The polynomial regression model captures this upward trend and forecasts nonlinear growth in energy consumption by 2030. This emphasizes the significant energy demand growth as data centers expand in scale and complexity.

Overall, we estimate total operational consumption(E_{op}) of HPC by combining the energy consumption of data centers and cryptocurrency, as shown in Equation 6, where E_{op} , $E_{d.c.}$ and E_{cry} denotes total operational energy consumption and energy consumed by data center and cryptocurrency.

$$E_{op} = E_{d.c.} + E_{cry} \quad (6)$$

As can be seen from Figure 4, data center energy consumption (purple line) has maintained a steady upward

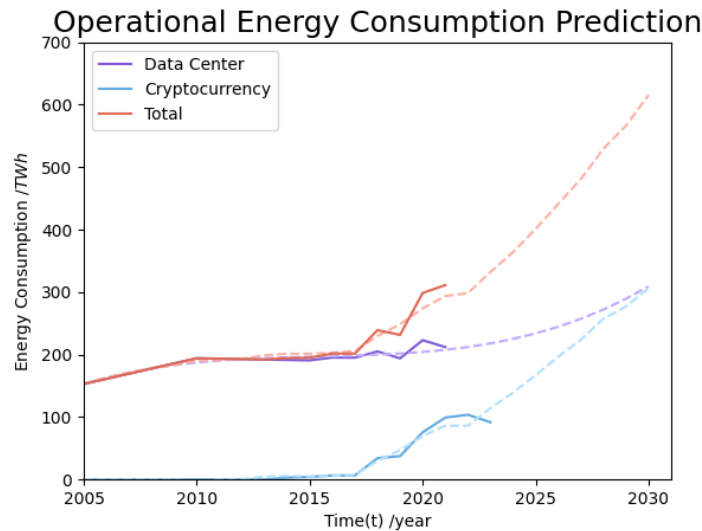


Figure 4: Global electricity consumption from data centers and cryptocurrencies prediction, 2005–2030

trend between 2005 and 2020, showing the continued growth of traditional computing and storage demand. In contrast, the energy consumption of cryptocurrency mining (blue line) has been rising rapidly since 2015, with a steeper growth trajectory than data centers and more volatile historical data.

Total energy consumption (orange line) shows the superimposed trend of data center and cryptocurrency mining energy demand. After 2020, a steep rise in the forecast curve suggests that the popularity of compute-intensive tasks will become an important driver of energy demand. In the future, with the wide application of

high-performance computing in scientific research, artificial intelligence and cloud services, the accelerated growth of its energy consumption is closely related to the rapid development of high-performance computing technology.

3.3.2 Energy resources

Since different energy sources contribute to the operational carbon footprint, we decided to find the total carbon emission by considering the amount of carbon emissions from using various energy sources and their proportion in energy consumption.

Carbon intensity refers to the amount of carbon dioxide (CO_2) emissions produced per unit of output. For each energy type, the carbon intensity ($c_{e.t.}$) varies. The data on carbon intensity is shown in Figure 5.

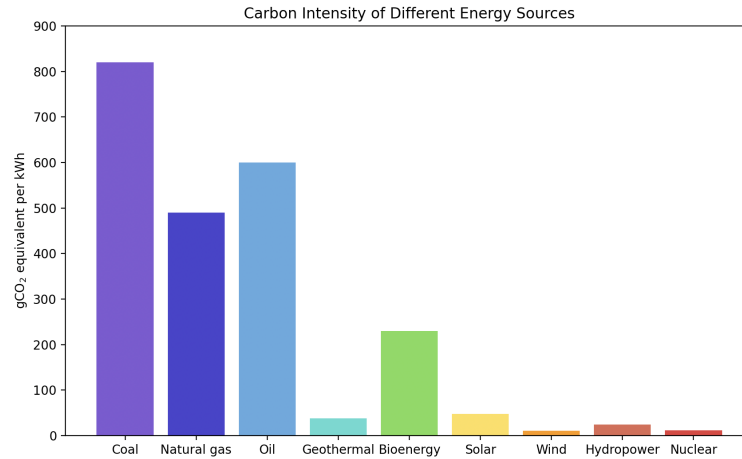


Figure 5: Carbon dioxide emission per kWh

Figure 5 presents the carbon intensity of different energy sources. The data shows that coal is the most carbon-intensive energy source, emitting $820gCO_2/kWh$, followed by oil and natural gas with emissions of $600gCO_2/kWh$ and $490gCO_2/kWh$, respectively. These fossil fuels are significant contributors to greenhouse gas emissions, emphasizing their detrimental environmental effects.

In contrast, renewable energy sources, including geothermal, solar, wind, and hydropower, along with nuclear energy, exhibit near-zero carbon emissions. This reflects their potential to minimize carbon footprints and their role as sustainable alternatives to fossil fuels. Bioenergy shows moderate carbon intensity, positioned between fossil fuels and renewable energies, suggesting a partial reliance on carbon-releasing processes.

For energy structure, we use polynomial regression model to predict the proportion of these sources globally, and the visualized forecast is presented in Figure 6 and Figure 7. [8]

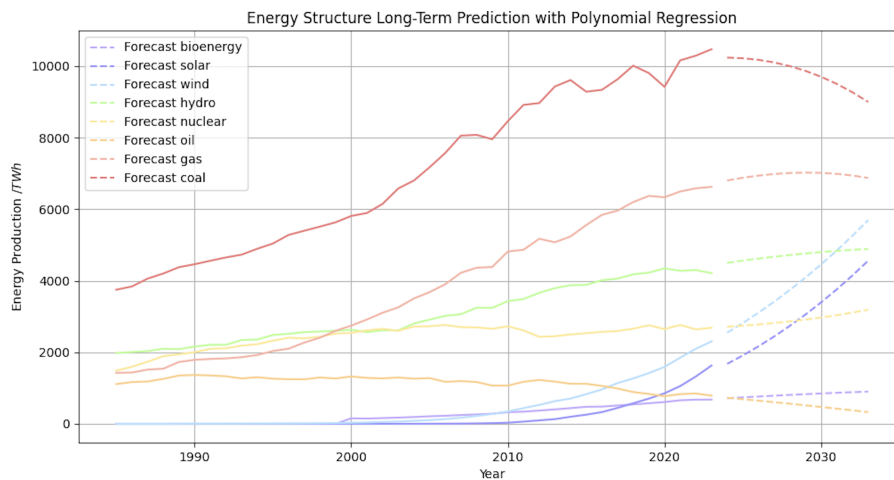


Figure 6: Global Energy Production Prediction

As shown in Figure 6, a polynomial regression model with past data, input is utilized to forecast the global energy production from eight independent energy sources, measured in TWh . The energy structure is then obtained by computing the percentage of energy occupied by each energy source, giving a visualized result in Figure 7. Historically, fossil fuels including coal, oil, and natural gas have dominated the energy

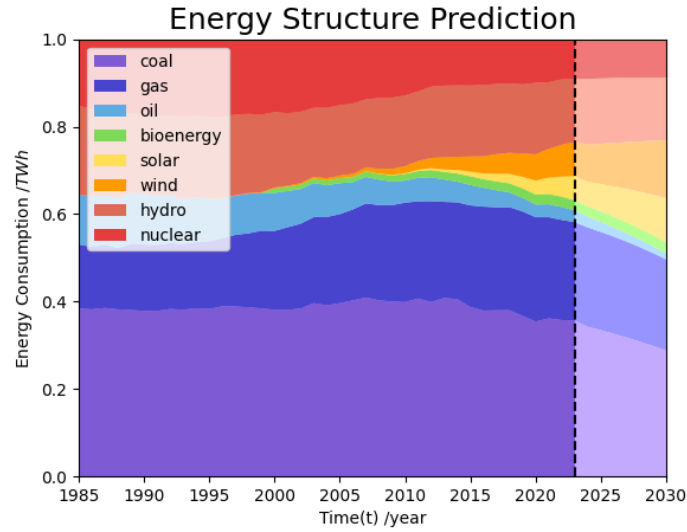


Figure 7: Global energy structure prediction

structure, occupying a large percentage of the global electricity supply. However, as time went on, we witnessed a gradual shift away from these traditional sources. In the predicted period after 2024, we observe a downward trend in fossil fuel use, while green energy sources such as wind, solar, and bioenergy exhibit an upward trend, occupying more positions in the global market.

The electricity demand for a particular energy source, in Equation 7, is the product of total operational energy demand and its proportion p in the world.

$$E_{e.t.} = E_{op} \cdot p \quad (7)$$

Government policies, technological developments and geographical factors have all influenced the evolution of the energy mix. By enacting energy policies and environmental regulations, governments can promote the use of clean energy or limit the development of fossil fuels. Scientific and technological progress determines the development efficiency and cost of new energy, such as solar energy, and battery technology breakthroughs to promote the popularity of renewable energy. Geographical conditions affect the distribution and utilization of energy resources, for example, solar energy is more advantageous in areas with sufficient sunshine, while wind energy is suitable for areas with abundant wind resources. The interaction of these factors determines the choice and adjustment direction of energy structure in a region. These variables add complexity to energy planning and are difficult to predict. However, they have a great effect on the optimization and sustainable development of energy structures and should be fully considered by people and the government.

C_{op} may be calculated using previous data and Equation 2. Figure 8 illustrates the projected trend in operational carbon emissions (C_{op}) from 2010 to 2030, measured in million tonnes of CO_2 [3].

Figure 8 demonstrates a steady increase in carbon emissions over the two decades, reflecting the rising energy demands of operational processes and the continued reliance on fossil fuels in many sectors.

From 2010 to 2020, the growth in emissions remains relatively moderate, suggesting the influence of energy efficiency improvements and initial efforts to adopt renewable energy sources. However, after 2020, a more pronounced upward trend is observed, indicating accelerated carbon emissions driven by increasing computational workloads, data center operations, and high-performance computing applications.

3.4 Infrastructure

Apart from direct ways of incurring increasing carbon emissions from electricity production. Greenhouse gases emitted from building and constructing facilities and devices for high-powered computing needs

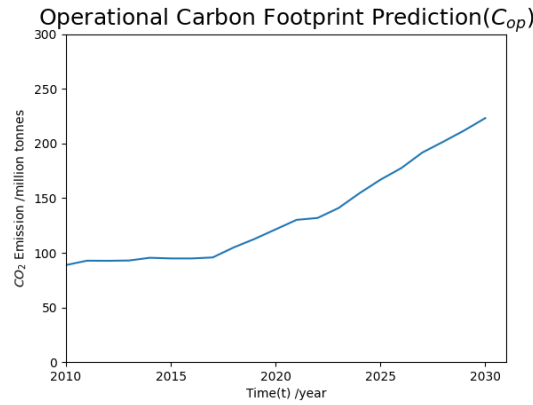


Figure 8: Operational Carbon Footprint Prediction

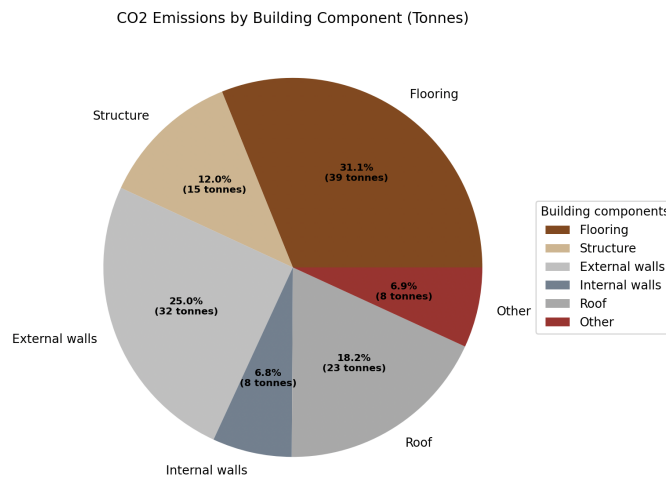
also play a significant role. Data centers require enormous quantities of concrete, steel, and other materials. These materials are produced through carbon-intensive processes. For instance, concrete involves cement production, which emits CO_2 during calcination (the conversion of limestone to lime). Also, Steel production uses coal-fired blast furnaces, releasing CO_2 .

This model will detail the carbon footprint associated with the infrastructure, breaking down the key stages and calculations. Through the model, we will evaluate the amount of carbon emission caused by the construction and development of infrastructure.

3.4.1 Carbon emission of construction

To estimate total CO_2 emissions, we estimate building carbon emissions for individual data centers. Dennis Bouley provided a method to calculate the infrastructure carbon footprint in his research. [7] [2] [9]

As shown in Figure 9, the pie chart illustrates the distribution of carbon emissions by an office building components for a 5700 ft^2 building, categorized into six main parts: flooring, structure, external walls, internal walls, roof, and other. [10]

Figure 9: Carbon emission by building components of a 5700 ft^2 building

Flooring contributes the highest share of CO_2 emissions, accounting for 31.1% (39 tonnes), followed by external walls at 25.0% (32 tonnes). The roof accounts for 18.2% (23 tonnes), while the structure contributes 12.0% (15 tonnes). Internal walls and other components have relatively smaller shares, both contributing approximately 6.8% and 6.9% (8 tonnes each), respectively. The average carbon footprint of concrete for an office building can be estimated as:

$$\frac{CO_2 \text{ of office building}}{5700 \text{ } ft^2} \times \text{Area} = 2250.88 \text{ tonnes}$$

The result shows us that an average of 2250.88 tonnes of carbon dioxide is estimated to be emitted from a data center of 100,000 square feet. However, that data center pollutes much more CO_2 during construction, because the infrastructure of the cooling system and operational system in the data center is highly carbon intensive and densely packed. The survey from APC states that data center emit 40 times more carbon dioxide than common office buildings. [7] Therefore, Equation 8 can be used to calculate infrastructure carbon footprint, where P denotes number of newly constructed data center every year.

$$C_{inf} = 2250.88 \cdot 40 \cdot P = 90035.2 \cdot P \quad (8)$$

3.4.2 Data center number

We define P as the increment of the data center each year because infrastructure construction requires one only for each data center.

To predict the trend of variable P over time, we decided to approach this question by finding a trend of hyper-scale data center numbers around the world. [12] Hyper-scale data center is a massive, highly scalable facility designed to support extensive digital sere. The number of hyper-scale data centers was estimated to be 28% due to ABIResearch. [13]

From the data, we come up with Equation 9.

$$E_{data\ center} = \frac{E_{hyperscale}}{28\%} \quad (9)$$

A visualized result of data center number increment is displayed at Figure 10.

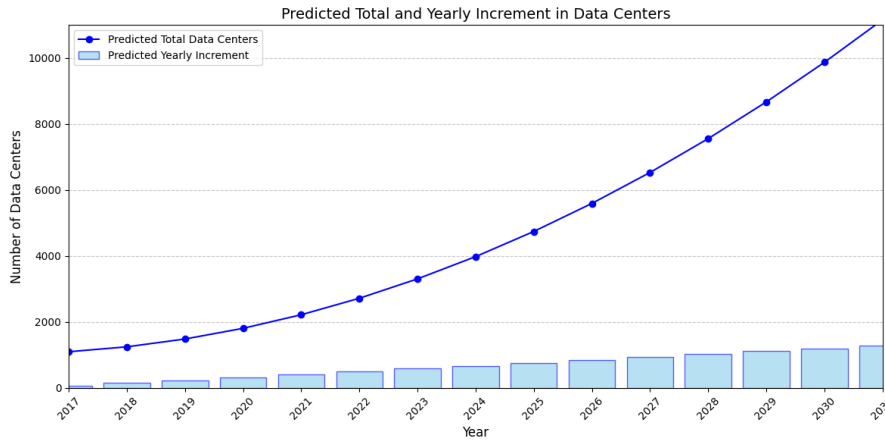


Figure 10: Predicted total and yearly increment in data centers

Figure 10 illustrates the predicted growth in the total number of data centers and their yearly increments from 2017 to 2031. The blue curve represents the total predicted number of data centers, while the blue bars indicate the yearly increments.

The total number of data centers is projected to grow exponentially, reaching over 10,000 by 2031. Yearly increments, shown as bars, also increase steadily but at a slower pace compared to the overall growth.

Following the result of Equation 8 and Figure 10, we predict the number of data center in 2030 using linear regression, because there exists a strong correlation between Time/year and Carbon footprint.

Figure 11 illustrates the predicted carbon footprint of infrastructure. The horizontal axis represents time (in years), while the vertical axis shows the carbon footprint in million tonnes. As observed, the carbon footprint exhibits a linear growth trend, indicating a consistent increase over time. The line represents the predicted yearly carbon footprint, starting near zero in 2018 and rising steadily to approximately 113 million tonnes by 2030.

3.5 Result

According to our previous calculation, we will obtain a carbon footprint prediction as shown in Figure 12. In the figure, we use the blue part to indicate the operational carbon emission, while the yellow region is used to present the infrastructure carbon emission. It shows a trend that, as time goes on, both operational

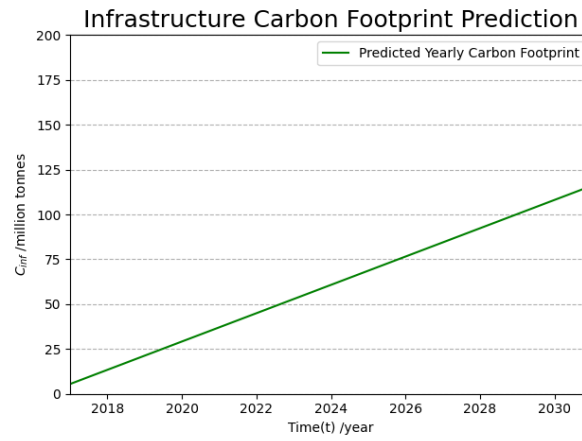


Figure 11: Infrastructure Carbon Footprint Prediction

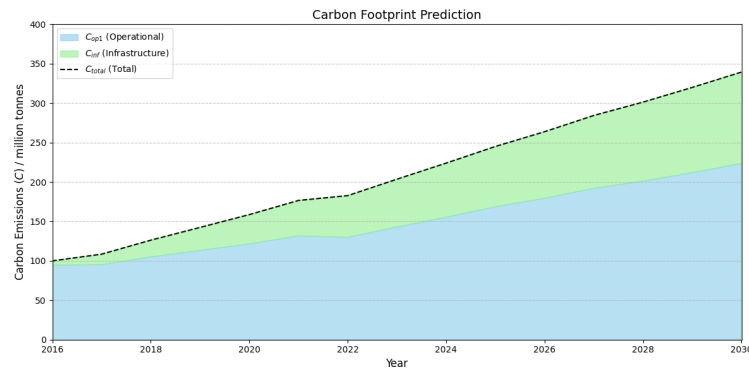


Figure 12: Carbon Footprint Prediction

carbon emissions and infrastructure carbon emissions increase. And the sum of them will reach about 340 million tonnes by the year 2030.

3.6 Prediction for year 2030

In the recent years, there exists a notable trend of demand for high-powered computing, driven by advancements in technology and a greater reliance on data analyzing. This surge in demand exerts a pulling force that is dragging the carbon dioxide emission rate higher year by year. A comparison between energy structure in 2024 and foreseeable 2030 is shown by two pie charts in Figure 15. It is noticeable that in both pie charts, non-renewable energy sources dominate the role of generating electricity, suggesting that their position remains stable, despite being 6 years from now.

While non-renewable energy sources such as coal, natural gas, and oil retain their primary position, their relative shares will fall by 2030. This decline can be attributed to several factors, including government policies, rising costs of fossil fuel extraction, and the increasing popularity of renewable energy sources.

Conversely, renewable energy sources, including hydroelectric, wind, nuclear, solar, and bioenergy, are expected to experience a notable growth in their contribution to the energy mix. This trend signifies a phenomenon of growing recognition of cleaner and more sustainable energy sources.

It is worth noting that nuclear energy, considered a low-carbon source, is projected to see a decrease in its share. The slow pace of nuclear power plant construction and technological innovation may contribute to this trend.

Furthermore, this result also inspires us to explore the potential consequences when this trend continues, which might be a prospect of the future. As renewable energy sources continue to gain shares of total energy and non-renewable sources gradually decline, it is imperative to consider the implications for energy security, economic stability, and environmental sustainability. We will explore it deeper in the next section.

In Figure 12, we predicted the total carbon footprint in the future, and the data of 2024 and 2030 are extracted in Figure 16 for a closer examination.

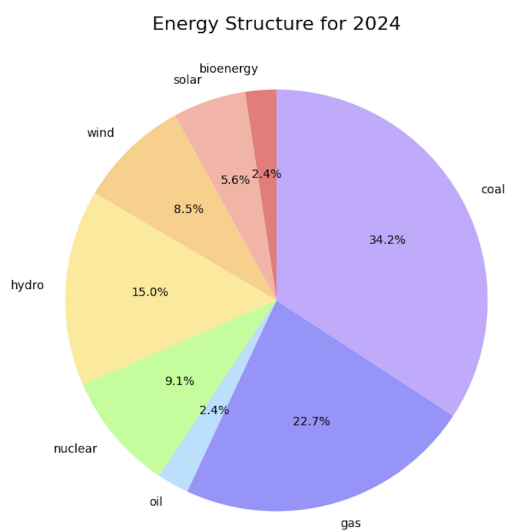


Figure 13: Energy Structure for 2024

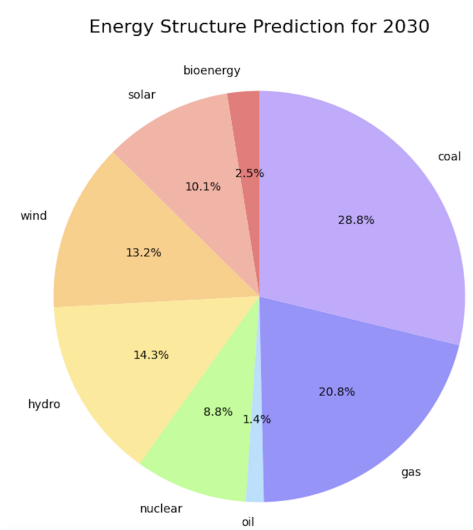


Figure 14: Energy Structure for 2030

Figure 15: Change of energy structure between 2024 and 2030

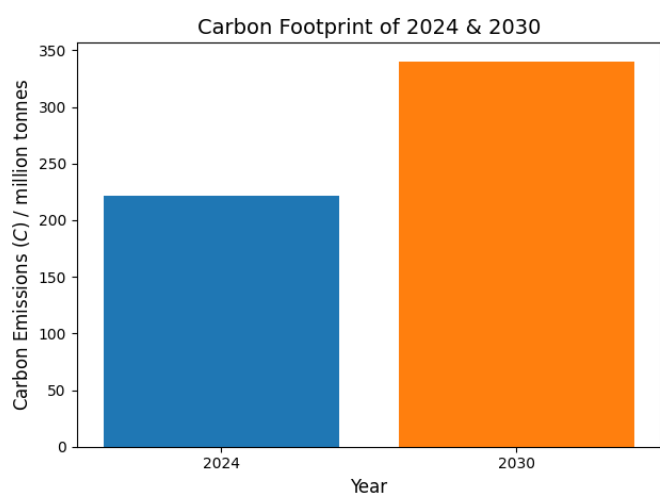


Figure 16: Carbon Footprint of 2024 compare to 2030

The bar chart compares the carbon emissions of infrastructure in 2024 and 2030, supported by precise numerical data. The blue bar represents the carbon footprint in 2024, which is 221.68 million tonnes, while the orange bar represents the footprint in 2030, measured at 339.66 million tonnes.

This indicates an absolute increase of approximately 117.98 million tonnes over six years, equivalent to a growth rate of about 53.2%. The significant rise reflects the continued expansion and energy demands of infrastructure development without adequate mitigation measures.

4 Effects of using renewable sources

Upon analyzing the data presented in Figure 7, we discovered that although green energy is conducive to the environment, it also has downside. Transitioning to a complete green energy system may induce various drawbacks that require thorough evaluation, and we must delve deeper into the implications of this transition to develop a more comprehensive and accurate prediction. To achieve this, we have chosen to focus on three critical aspects: carbon dioxide emissions, land consumption, and costs.

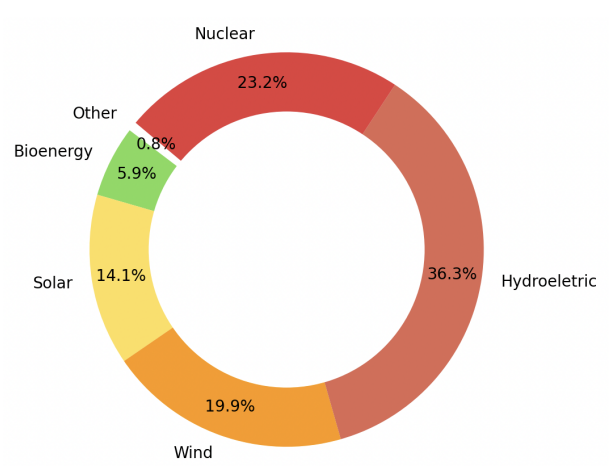


Figure 17: Bar chart of global renewable energy proportions

4.1 Emission of CO₂

The carbon dioxide emission is affected by the carbon intensity(*c*). The value of *c* for individual energy sources is different because the chemical or physical method they use to generate electricity is different, and produces CO₂ in different rates. It is noticeable that using more renewable energy sources will reduce the carbon emission of the data center in the operational aspect, but the emission in the infrastructure aspect will remain the same. A graph of CO₂ emission comparison for the current situation and hypothetical 100% renewable source is displayed in Figure 18.

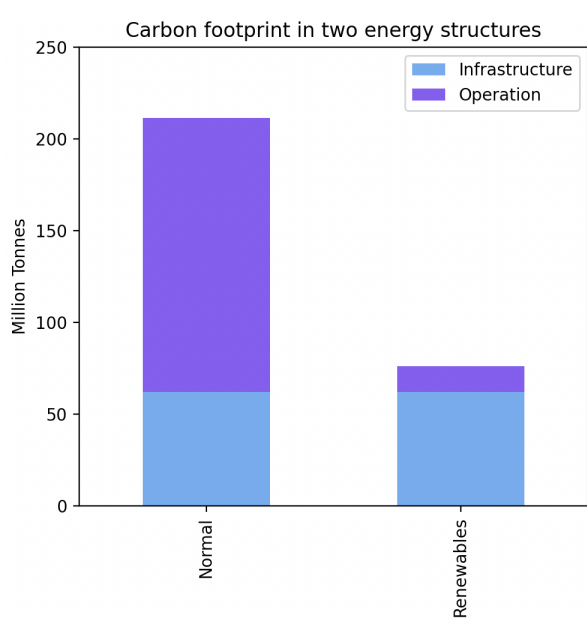


Figure 18: Carbon footprint of two energy energy structure, 2024

In Figure 18, we discovered that the difference in carbon footprint in two energy structures is 149 million dollars because using 100% renewable energy will significantly reduce the operational carbon down to less than 10%. We may deduce that a combination of high renewable source proportion and low increment of the data center is most conducive to the environment. The switching towards renewable energy is most effective when infrastructure carbon footprint is low.

This implies that determining the balance between renewable energy adoption and infrastructure development are crucial for achieving sustainability goals. For regions with existing infrastructure that is energy-intensive, the transition to renewable energy sources should be prioritized. Conversely, for regions in rapid development, careful planning of data center growth is vital.

4.2 Land consumption

Despite less greenhouse gas pollution, 100% renewable energy sources will consume more area than normal to build the infrastructure to produce the same amount of energy, especially for wind and hydro-electric. Renewable energy generation occupies a large amount of area that was originally available for agricultural or residential use. Bioenergy uses $45m^2$ of land per MWh of electricity generated, nearly twice as much as solar, because it converts organic materials such as plant materials, and agricultural and forestry residues, into electricity. Detailed data is shown in Figure 19.

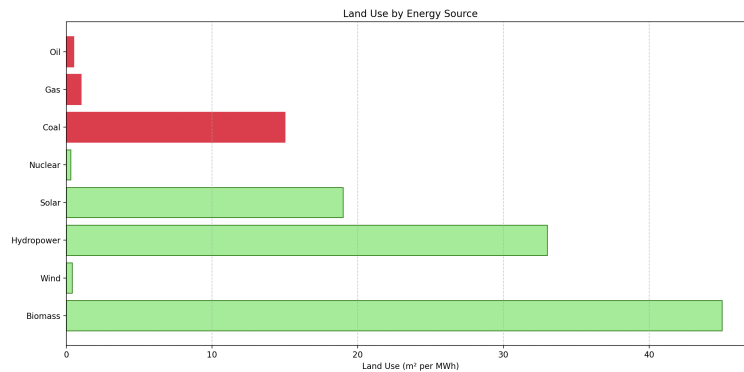


Figure 19: Land consumption of different energy sources

With data from Figure 19, global energy proportion in Figure 13, and energy consumption of data center in Figure 3, it is straightforward to predict the land use in acre when using 100% renewable, compared with normal conditions in 2024. The results is displayed in Figure 20.

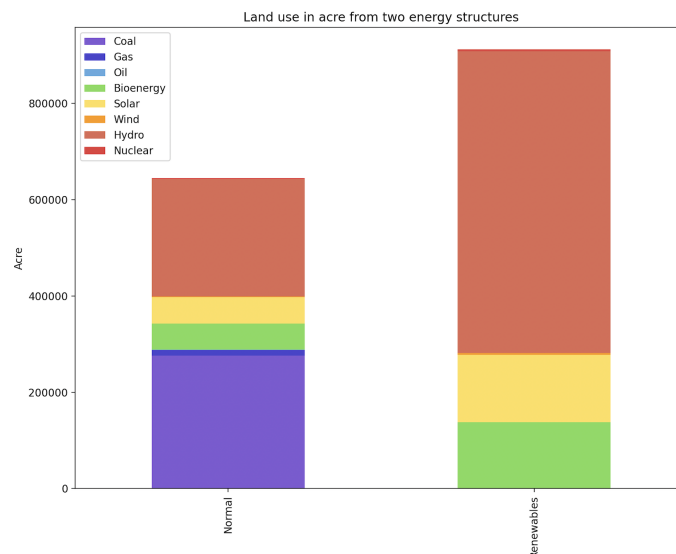


Figure 20: Land in acre for 100% renewable compared with current condition

In Figure 20, we compared the land use of two energy structures measured in acre, and concluded that switching to 100% renewable energy source implicates 267 thousand acre more area. This rise in land use is primarily caused by hydroelectric power which requires $33m^2/MWh$, and shares a portion of 36.3% among renewables. Therefore, hydroelectric energy is not an optimum energy source for regions with strict land resources.

On the other hand, wind and nuclear energies are encouraged for small regions, as they are both environmentally and land-effective. Moreover, coal is dissuaded, since it both pollutes large amounts of CO_2 and requires extensive land.

4.3 Lower Cost

Although renewable energy sources seem to require high costs, most of the expenditures come from infrastructures, and the average operational cost actually is lower than the current condition. This is because renewable energy such as solar panels and wind turbines generate electricity from the natural environment, which is a free good. Compared with oil, coal and gas which involves the excavation of raw materials and fuels, renewable energy has an inherent advantage. Detailed data on energy source cost is shown in Figure 21.

Also, we researched the structure of renewable energy sources from 2013 to 2023, visualized in Figure 17. By combining them, we can calculate the cost of the same figure of electricity when we use 100% renewable energy source, compared with normal conditions.

The result of our model is shown in Figure 22, illustrating the cost of money from HPC use when using 100% renewable, compared with normal conditions in 2024.

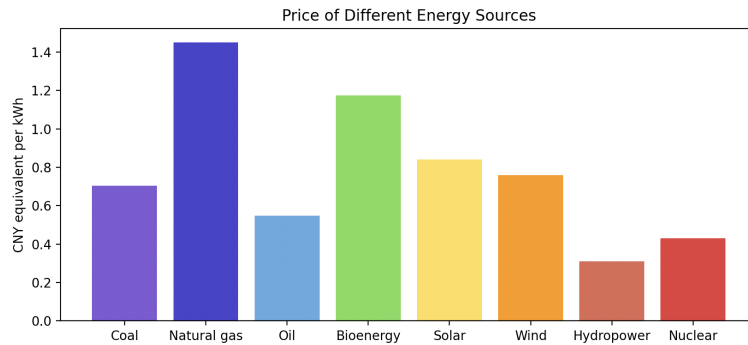


Figure 21: Price of different energy sources

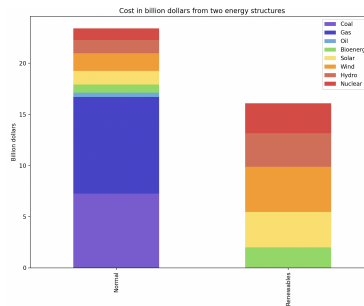


Figure 22: Cost in billion dollars for 100% renewable compared with current condition

In Figure 22, we compared the costs associated with two energy structures, measured in billion dollars. The analysis reveals that transitioning to a 100% renewable energy scenario would reduce a cost of 7.32 billion dollars. This significant reduction is driven by the lower costs of hydroelectric and nuclear power.

Despite the higher initial investment, renewable energy offers long-term economic benefits through reduced fuel costs and environmental advantages. Notably, nuclear energy presents a unique case, as nuclear power plants are expensive but offer a low maintenance price and reliable energy sources. Although natural gas and oil are easily acquired, the cost of excavation is substantial, which is disadvantageous in the long term.

5 Effects on Network Architecture

The network infrastructure needed for data transmission extends beyond individual nodes to include greater connectivity concerns. As the development of HPC, the number of data center increases.

We will explore the relationship between the number of data centers and the cost to build nodes, the cost for data transmission, traffic during data transmission, the average number of nodes each data package travels and the possibility the nodes break during transmission. By using linear programming, the optimal placement of data centers can be determined, balancing the trade-off between cost and latency. Then, we

will use Dijkstra's algorithm to assess the network's overall traffic and energy to ensure the most efficient data transmission routes.

5.1 Graph

The network is represented as a graph, where the vertices represent nodes and the edges represent the distance between connected routers.

In the model, we assume that each node will host user requests in its vicinity. The nodes are also alternate addresses for the data center, which are crucial hubs for processing and distributing data. Once the location of the data center is selected, the normal nodes can only transmit data to the data center nearest to it. Data centers can transmit data to each other. This graph helps in understanding the topology and how data moves across the network.

We use Figure 23 below to visualize the network structure of Amazon's data center as an example. [5] [11] We will use the same graph to test our model. Normal nodes are presented in red while data centers are in green.

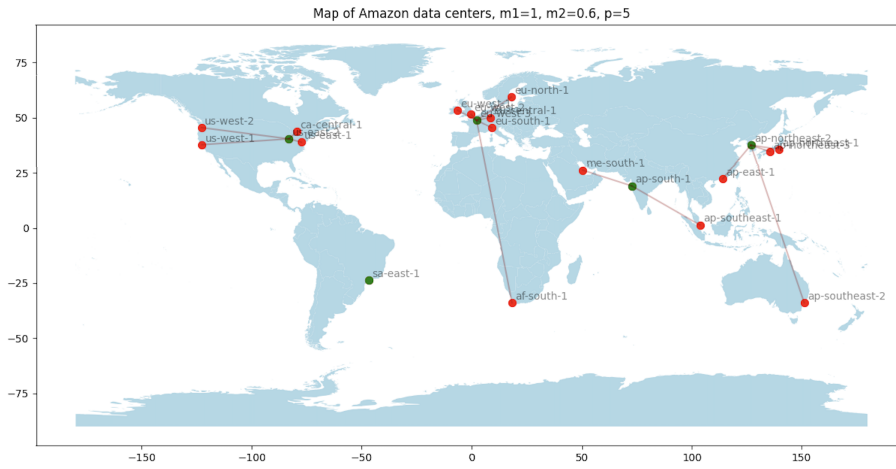


Figure 23: Sample structure of network

5.2 Data center location

Entrepreneurs need to determine where to place the data centers. In our model, the two primary factors affecting this decision are considered.

First of all, the construction cost. We are going to determine how expensive it is to build a data center at a particular node. In addition, the service quality is indicated by latency. It represents how quickly data can travel between nodes.

We deploy an Integer Linear Programming (ILP) model to decide the optimal locations for data centers, which is shown in Equation 10 and Equation 11. In this model, we aim to minimize a cost function that balances latency and construction costs.

$$\min m_1 \cdot \sum_i \sum_j L_{i,j} \cdot Y_{i,j} \cdot D_i + m_2 \cdot \sum_i X_i \cdot K_i \quad (10)$$

$$s.t. \begin{cases} \sum_i X_i = p \\ \sum_i Y_{i,j} = 1 \\ Y_{i,j} \leq X_i \\ X_i \in \{0, 1\} \\ Y_{i,j} \in \{0, 1\} \\ \forall i, j \in \mathbf{F} \end{cases} \quad (11)$$

In our model, we tend to achieve our aim by minimizing the weighted sum of the total latency across the network based on the traffic demand between nodes and the total construction cost for placing data centers at specific nodes. [6] We add several constraints: We use $X_i \in \{0, 1\}$ to ensure that a data center is either

placed at node i or not. $Y_{i,j} \in \{0, 1\}$ is used to ensure that each node j is served by exactly one data center. We define P as the number of data centers. And a node i can only serve node j if a data center is placed at node i , which is captured by $Y_{i,j} \leq X_i$.

As ILP problems are NP-hard, we use a heuristic method here to approximate the global optimization in a large search space, which is called Simulated Annealing (SA). Its pseudo-code implementation is shown in Algorithm 1.

In lines 3 - 6, we initialize state X and Y with random binary values (0 or 1), and calculate the obj using them, then X_{best} , Y_{best} and obj_{best} , are set to the same value of those variables.

Lines 10 - 24 are the loop of simulated annealing iterations.

New neighbor state, X_{new} and Y_{new} , which means randomly flip one binary from X or Y , is selected at line 11, and the obj_{new} is calculated at line 12.

In lines 13 - 18, we accept the new state X_{new} and Y_{new} when obj_{new} is better than obj (where better means less here), or randomly accept it when it is worse, at a probability decreasing with T .

In lines 19 - 22, the best state is updated if new state has a less objective value. Function `UpdateStateIfBetter` here is a placeholder for a if statement to do the conditional update.

Finally, T is set to $T \cdot cooling_rate$, so temperature goes lower.

Algorithm 1 Simulated Annealing

```

1: Input:  $n, p, L, D, K, m_1, m_2$ 
2: Output: Best solution  $X, Y$ , and its objective value
3:  $X \leftarrow \text{RandomBinaryArray}(n)$ 
4:  $Y \leftarrow \text{RandomBinaryMatrix}(n, n)$ 
5:  $obj \leftarrow m_1 \cdot \sum(L \cdot Y \cdot D) + m_2 \cdot \sum(X \cdot K)$ 
6:  $X_{best}, Y_{best}, obj_{best} \leftarrow X, Y, obj$ 
7:  $T \leftarrow \text{InitialTemperature}$  ▷ Initial temperature, e.g., 1000
8:  $cooling\_rate \leftarrow 0.99$ 
9:  $max\_iter \leftarrow 10000$ 
10: for  $i = 1$  to  $max\_iter$  do
11:    $X_{new}, Y_{new} \leftarrow \text{RandomNeighborStateOf}(X, Y)$ 
12:    $obj_{new} \leftarrow m_1 \cdot \sum(L \cdot Y_{new} \cdot D) + m_2 \cdot \sum(X_{new} \cdot K)$ 
13:   if  $obj_{new} < obj$  then
14:      $accept \leftarrow \text{True}$  ▷ Always accept better solutions
15:   else
16:      $\Delta \leftarrow obj_{new} - obj$ 
17:      $accept \leftarrow \text{Random}() < \exp(-\Delta/T)$  ▷ Accept worse with a probability decreasing with  $T$ 
18:   end if
19:   if  $accept$  then
20:      $X, Y, obj \leftarrow X_{new}, Y_{new}, obj_{new}$ 
21:      $\text{UpdateStateIfBetter}(X, Y)$ 
22:   end if
23:    $T \leftarrow T \cdot cooling\_rate$ 
24: end for
25: return  $X_{best}, Y_{best}, obj_{best}$ 

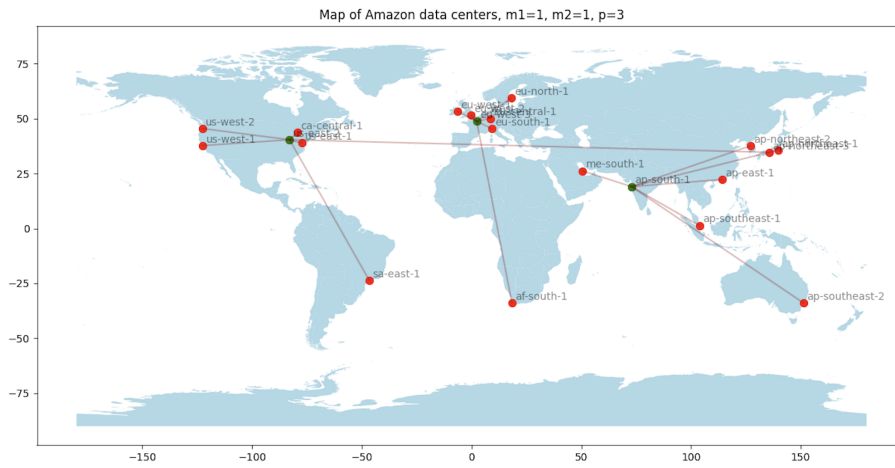
```

We use $m_1 = 1$, $m_2 = 1$ to simulate the preference of companies, meaning that they think both cost and service quality are of the same importance.

As the result of linear programming, the best site is when $P = 3$, as shown in Figure 24.

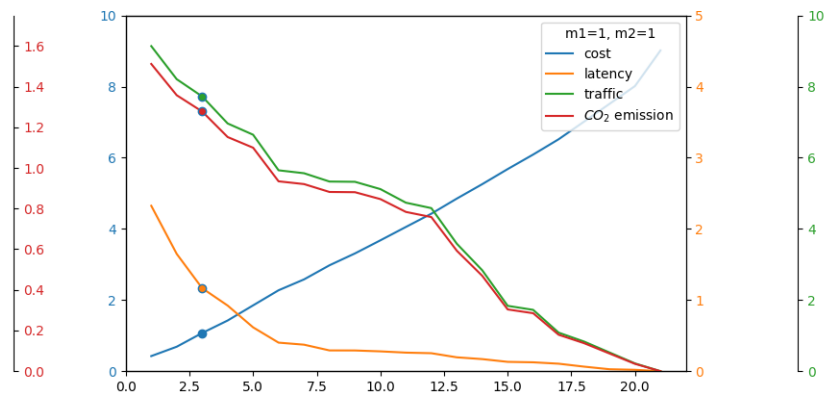
5.3 Traffic

In real life, the traffic load will influence network performance, as higher traffic means higher energy consumption. According to a study from Global Energy Interconnection Development and Cooperation Organizations, the global data center traffic will reach 60,000 EB in 2030. Therefore, a large amount of energy is needed for enormous traffic. Also, the placement of data centers and the routing decisions will affect the overall network traffic. For example, placing a data center at a particular location could reduce latency but might increase traffic for certain nodes, leading to more energy consumption. Conversely, if a data center is placed further from high-traffic nodes, it may increase latency but reduce energy consumption.

Figure 24: Map of Amazon data centers, $m_2 = 1$

To explore how data travels across the network, we will use Dijkstra's algorithm to find the shortest path for data transmission between nodes. Since the Dijkstra's algorithm is a classic method for determining the shortest paths in a graph, we can obtain efficient data routing in this way.

Different location will cause different traffic as shown in Figure 25

Figure 25: When $m_2=1.0$

The results of Dijkstra and Linear Programming models are presented in Figure 25, where the x-axis represents the number of data centers connected to the new one. The blue line represents the cost function which will increase with the number of data centers, as a higher complexity of network connection will increase the rate of data transmission and one bit of data transmitted costs $0.2mJ$ of energy. [14] The orange line indicates the latency index that declines gradually as data center connections increase because more direct connections are available. The green and red lines refer to traffic and CO_2 emission function respectively, which are strongly correlated.

Our result is signified by four dots, where the linear programming function is minimized and the number of connected data centers is 3. This result implies that too many or too few data center connected are both averse to efficiency.

6 Site selection to reduce the carbon footprint

Optimizing the location of data centers can reduce the energy consumption caused by traffic, and the energy mix of different regions can lead to different CO_2 emissions. In response to this situation, the government can give subsidies to companies to optimize the location of data centers, to achieve the purpose of carbon dioxide reduction.

Taking CO_2 emissions and network latency as priorities, we use linear programming(the same model in 5.2) to simulate the optimal location for a given number of data centers p . We modulated the preference priority latency by decreasing the value of m_2 under the government’s subsidies. By calculation, the new model siting reduces carbon dioxide emission than the old model.

We find out the situation when $m_2 = 0.8$, $m_2 = 0.6$, and $m_2 = 0.1$ as shown below.

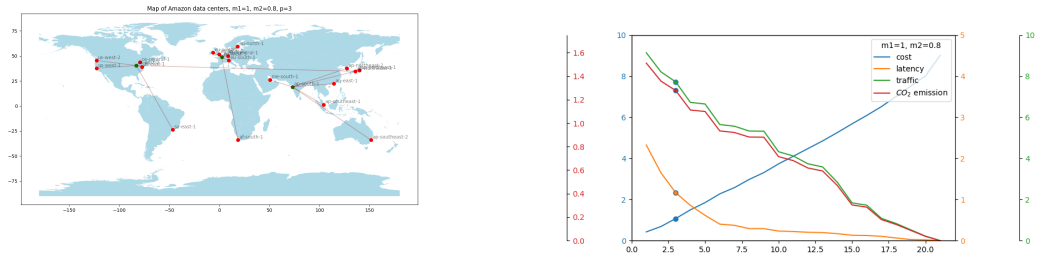


Figure 26: Map of data centers, $m_2 = 0.8$

By running the program, we find that the impact of government subsidies for data center location on CO_2 emissions is shown in Figure 29.

This chart illustrates the relationship between the Carbon Emission Index and the Portion of Government Subsidies. The horizontal axis represents the proportion of government subsidies (ranging from 0 to 1), while the vertical axis shows the normalized value of the Carbon Emission Index (ranging from 0 to 1), providing a clear view of the relative trends.

The chart reveals that as the proportion of government subsidies increases, the Carbon Emission Index decreases progressively. This indicates that increased subsidies may drive the adoption of more environmentally friendly technologies or industrial transformations, thereby reducing overall carbon emissions. At lower subsidy levels (0.0 to around 0.2), the Carbon Emission Index remains nearly constant, suggesting that small amounts of subsidies might be insufficient to significantly impact carbon emissions. When the subsidy proportion exceeds a certain threshold (around 0.2), the Carbon Emission Index begins to show a noticeable decline, indicating that subsidies in this range likely trigger broader green transformation efforts.

Furthermore, as the subsidy proportion approaches 1.0, the Carbon Emission Index drops sharply, nearing zero. This may reflect the cumulative and nonlinear effects of higher subsidies on emission reductions. Overall, this chart clearly demonstrates the importance of government subsidies in promoting emission reduction. It also provides quantitative insights for policymakers, showing that allocating subsidies appropriately can more effectively achieve emission reduction goals.

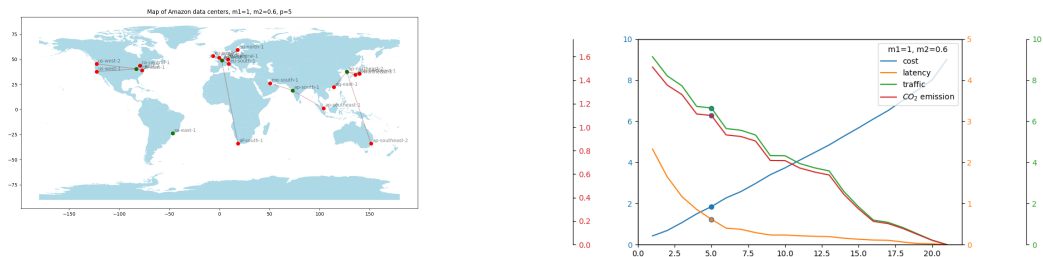


Figure 27: Map of data centers, $m_2 = 0.6$

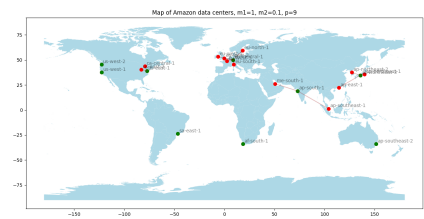
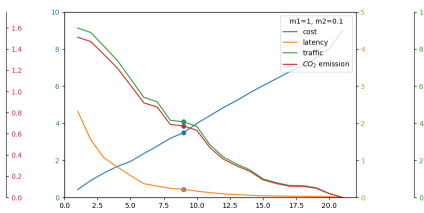


Figure 28: Map of Amazon data centers, $m_2 = 0.1$



7 Conclusion

7.1 Recommendations

Base on our results, we tend to develop and offer a mix of technical and policy-oriented solutions to reduce the environmental impact of High-Performance Computing (HPC). We will basically divide our suggestion into 4 points:

Recommendation 1 Promote Energy-Efficient HPC Technologies.
As government and institutions can play pivotal role in fostering innovation for energy-efficient HPC systems, we can promote the development of technologies. For example, supporting the use of low-power, high-performance computing chips and energy-aware hardware design can help reduce overall electricity consumption in data centers. Incentives could be recommended for organizations that research and deploy energy-efficient hardware solutions in HPC. These grants and subsidies can accelerate adoption and innovation of technologies, so that overall energy consumption in data centers can be reduced.

Recommendation 2 Implement Carbon Tracking and Accountability for HPC.
Transparency and accountability are essential to measure and mitigate the environmental impact of HPC. Government can achieve it by develop industry-wide protocols for measuring and reporting the carbon

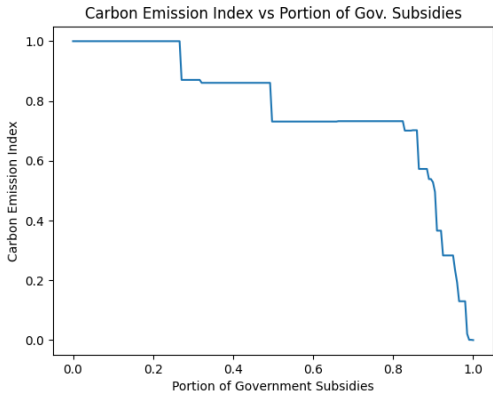


Figure 29: Carbon Emission Index vs Portion of Gov. Subsidies

footprint of HPC activities ensures consistency and comparability. Organization can also publish reports detailing their energy usage and associated emissions builds accountability and highlights areas for improvement. This measures can ensure the achievement of sustainability goals by making data transparent.

Recommendation 3 Encourage the use of Renewable Energy in HPC Facilities.

We encourage transition to renewable energy sources, which can be seen as a vital step in reducing the carbon footprint of HPC facilities. Government can establish clear guidelines for the percentage of energy derived from renewable sources, such as solar, wind and hydropower. They can help build and integrate renewable energy sources into existing data centers. Also, organizations can be encouraged to use certified green energy by receiving incentive from government.

Recommendation 4 Support development of Green Network Architectures.

Since HPC systems rely on complex networks for data transfer, we can optimized for sustainability through energy-efficient network architecture. By using our models, we can design systems that minimize energy use during data transfers between computing nodes. Also, during data transferring, the traditional copper cables can be replaced by optical solutions, which can significantly lower energy requirements. Focusing on network sustainability can reduce the environmental impact of the large-scale data handling integral to HPC operations.

7.1.1 Implementation of recommendation

We assume that our recommendation 4, "supporting the development of Green Network Architecture", is acted upon.

According to Problem 4 and 5, when subsidies are given to companies, companies can optimize the location of data centers to achieve the purpose of carbon dioxide reduction.

We will then deploy the linear programming (LP) in 6.2 to determine the optimal locations for data centers. The model will output a site selection that not only reduces carbon dioxide emissions and latency but also includes the sustainable energy usage from the new network architecture, results in a more holistic and environmentally sustainable model for High-Performance computing (HPC) systems.

7.2 Strength

1. Comprehensive Modeling Approach: We use detailed models to predict carbon emissions, assess renewable energy impacts, and optimize network architectures.
2. Use of Advanced Prediction Techniques: We deploy time series models and linear programming showcases a methodical approach to address the problem.
3. Detailed Data Integration: Incorporates diverse datasets to present clear insights. For example, energy structure, land use, and cost analysis for renewable transitions.

7.3 Weaknesses

1. Limited Scope in Assumptions: Key factors like technological advancements and regional policies are excluded, which might affect the real-world applicability of results.
2. Focus on Specific Sectors: We only consider data center and cryptocurrency energy usage, potentially overlooking other significant HPC applications.
3. Geographical and Political Context Missing: The solutions lack customization for different regions, where energy policies and availability of renewable sources vary significantly.

8 Letter to United Nations Advisory Board

Dear Members of United Nations Advisory Board:

As students who are concerned about advanced technology and sustainable development of society, we all commend the board's release of the report "Governing AI for Humanity" and its proactive stance on AI governance and ethical responsibility, since this dedication to ensure AI serves humanity and respects human rights is critical in the path to a more prosperous future. However, We are writing to suggest the Advisory Board to consider an additional, vital aspect that was minimally addressed in the previous report: the environmental impact of High-Performance Computing (HPC) technologies. Given that High-Performance Computing's crucial role in AI model development and deployment, this omission presents a substantial gap in United Nation's agenda. Falling to address the environmental impacts of the use of High-Performance Computing could directly undermine the progress toward the UN's scheduled development goals for 2030 of Sustainable Development Goals (SDGs), particularly Goal 7 (Affordable and Clean energy) and Goal 13 (Climate Action).

Since the continually advancement of AI, High-Performance Computing systems have become essential for social development, we must consider the external costs caused by it: the significant energy consumption and carbon dioxide emissions. There is no shortage of evidence showing the massive environmental impact of AI development: For example, according to a recent article published in Nature, searches powered by generative AI consume four to five times more energy than traditional web searches, and ChatGPT, which is one of the most popular chat-bot nowadays created by OpenAI in San Francisco, California, has consumed the energy of 33,000 homes.

Therefore, it is urgent for the United Nations to include a more detailed section on the environmental impacts of HPC in the scheduled developmental goals for 2030. We propose the following recommendations for your consideration:

1. Promote Energy-Efficient HPC Technologies.

It is crucial for governments to encourage the development of energy-efficient processors and systems in HPC infrastructures. It can help reduce overall electricity consumption in data centers. Incentives could be recommended for organizations that research and deploy energy-efficient hardware solutions in HPC.

2. Implement Carbon Tracking and Accountability for HPC.

Establish standard practices for tracking and reporting carbon emissions specific to HPC activities to help create a transparency and accountability. Organizations that using HPC infrastructure could be recommended to publish annual carbon impact reports, which is focused on energy usage and carbon emissions linked to their computational workloads. This can enable policymakers and stakeholders to track progress toward sustainability goals.

3. Encourage the use of Renewable Energy in HPC Facilities.

Urge organizations and institutions relying on HPC to transition to renewable energy sources. It could involve setting guidelines or targets for clean energy use in data centers, such as solar, wind and hydropower.

- 4.Support development of Green Network Architectures.

Recommend the adoption of energy-efficient network architectures that minimize the environmental impact of data transfer in HPC systems. Energy-aware routing algorithms, optical interconnects, and data transfer protocols specifically designed for HPC can reduce the energy consumption and carbon emission significantly, making HPC networks more sustainable.

To assist the application, we develop a set of models to predict the carbon emission, explore the effect when renewable resources deployed and also the effects on network architecture. We hope that you find the flexibility and efficacy of our model compelling, and adopt it for future application.

Thank you for your attention to this important issue. We believe the unique position of United Nation as a global convener of nations and thought leaders, and we trust that your stance on HPC and its environmental impacts would carry substantial influence. We are confident that with the Advisory Board's guidance, we can harness the power of AI responsibly, without compromising environmental health.

Sincerely,
Team 15260

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